Olist: late deliveries are costing reviews — and the cheapest fix

Insights memo · 96,203 delivered orders, Jan 2017 – Aug 2018 · Brazilian E-Commerce Public Dataset by Olist (Kaggle) · Prepared by Ideraoluwa, Data Analyst

Bottom line

Only **6.8%** of orders arrive after the promised date, but they generate **37% of all 1-star reviews**. Customers judge the *promise*, not the speed: an order that takes 22–30 days still scores 3.95 if it meets its date and 2.57 if it misses it. Lateness is concentrated on a dozen routes and two periods, and in 72% of cases the seller shipped on time — the carrier leg is where the time is lost. The cheapest fix is a route-aware delivery promise; the second is a carrier SLA on Rio de Janeiro and the Northeast. Delivery is **not** a retention lever: 97% of customers never reorder regardless.

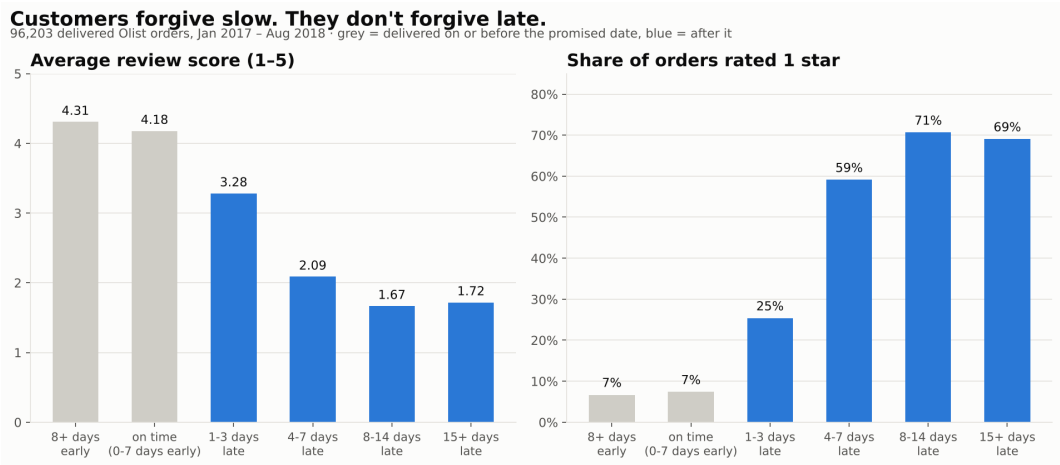
96,203	6.8%	4.28 vs 2.26	6.8% vs 54.2%	37%	3.0%
delivered orders	delivered late	avg review: on time vs late	1-star rate: on time vs late	of 1-star reviews come from late orders	of customers ever reorder

What the data says

- **A missed promise turns a 4.3-star experience into a 1-star one.** 1–3 days late: avg score 3.28, 25% 1-star. 4–7 days late: 2.09, 59% 1-star. Past a week: ~1.7, 70% 1-star. Only 6.8% of orders arrive late, but they produce 37% of all 1-star reviews (54% of late orders get 1 star vs 7% of on-time orders).
- **The promise matters more than the speed.** At every delivery-time band, orders that met the promised date score 0.5–1.8 stars higher than orders of the same speed that missed it. The estimate is padded by ~12 days on average, yet it is the *same* padding everywhere — so it fails exactly where transit is slow.
- **Lateness is concentrated.** 12 seller→customer routes produce 69% of late orders (SP→RJ alone 18%, late 14% of the time). Deliveries to Rio de Janeiro and the nine Northeast states are 22% of orders but 40% of late ones. The late rate spikes in Nov 2017 (12%) and Feb–Mar 2018 (14–19%). Late orders spend 28 days in carrier transit vs 8 for on-time orders; seller handling only grows from 2.6 to 5.5 days.
- **Multi-item orders are a second problem.** Orders with 2+ items get 1 star 2.5× as often (21.7% vs 8.7%), orders split across sellers 36% — although they are *less* often late. Consistent with split or partial deliveries.
- **Retention is structural.** 97.0% of customers never place a second order. A late first order lowers the repeat rate from 3.03% to 2.52% — real but small — and the first order's review score does not predict a return at all.

Recommendation, cheapest first

1. **Make the delivery promise route-aware.** A 7-day buffer on the 12 worst routes alone would have turned 40% of all late orders into on-time ones — ≈ 1,550 orders and ≈ 730 fewer 1-star reviews a year; with the Nov–Mar peak added on every route, 49% (≈ 900 reviews); applied everywhere, 56% (≈ 1,040), lifting the platform average from 4.14 to 4.22. Zero logistics cost — A/B test it against checkout conversion.
2. **Carrier SLA on deliveries to Rio de Janeiro and the Northeast.** Cutting the overall late share from 6.8% to 3.5% moves ≈ 1,900 orders a year on time and avoids ≈ 900 1-star reviews.
3. **Investigate split deliveries on multi-item orders** (1-star reviews 2.5× as often, 4.1× for multi-seller orders), and **stop measuring delivery as a retention lever** — measure it on review score and 1-star rate.



Method: MySQL 8 (schema, 11 analysis queries), Python (pandas, matplotlib), Power BI (5-page dashboard, star schema, DAX), Excel (what-if model). Late = delivered after the estimated date; revenue = item price; review = latest per order. Limits: review effects are correlational; carrier costs and Olist's take rate are not in the data. Full repository: github.com/[your-handle]/olist-retention-delivery-analysis